

Héctor Ramón Martínez de la Hoya

Data Engineer / Machine Learning Engineer

+52 618 234 3529
devlahoya@gmail.com

Professional Summary

Data Engineer and Machine Learning Engineer with a strong background in software development, distributed/HPC computing, and applied artificial intelligence. Experienced in designing scalable ML pipelines, engineering GPU-accelerated computational systems, and managing large-scale batch workloads on Linux/SLURM infrastructure across national and international supercomputing centers. Proven ability to translate technical requirements into production-ready solutions and collaborate in multidisciplinary, cross-border teams.

Professional Experience

- Aug 2026 – Dec 2026 **Research Assistant – Computational Systems (J-1 Research Scholar)**, *New Mexico State University*, Las Cruces, NM, USA
- Serving as Research Assistant to Dr. Efren Delgado, extending prior ML pipeline work to a new applied research setting.
 - Executing large-scale computational workflows on the Discovery supercomputing cluster, including job scheduling, resource allocation, and pipeline optimization in a Linux/HPC environment.
 - Collaborating with U.S.-based faculty and researchers in a cross-border technical team.
- Jan 2025 – Dec 2026 **Machine Learning Engineer – Applied Research**, *Durango Institute of Technology – National Technological Institute of Mexico*
- Designed and trained deep learning models (Python, PyTorch, Transformer-based architectures) as part of an end-to-end applied AI pipeline.
 - Engineered a custom GPU-accelerated computational engine from scratch (FFT-based, with checkpoint/resume support), outperforming existing reference tools by up to 20x in processing time.
 - Managed large-scale distributed computing jobs on the LNS-BUAP Huapáctic supercomputer, including SLURM job scheduling, queue optimization, and concurrency handling on Linux.
- Nov 2022 – Aug 2024 **React Developer | Project Manager**, *QACS*
- Led the development and optimization of web applications using React, focusing on performance, scalability, and user experience.
 - Worked across the full stack using JavaScript, HTML, and CSS, integrating solutions with CMS platforms such as WordPress (Elementor, Divi).
 - Managed cross-functional teams and project timelines, delivering high-quality digital solutions for private-sector and government clients.
- Aug 2019 – Sep 2022 **IT Systems Manager**, *Magdalena Construcciones*
- Managed and maintained enterprise IT infrastructure, including system deployment, upgrades, and ongoing support.
 - Strengthened network security and implemented best practices to protect sensitive and critical business data.
 - Developed and deployed internal software solutions to improve operational efficiency and workflow automation.

Education

- Jan 2025 – Dec 2026 **M.S. in Engineering (Systems Engineering)**, *Durango Institute of Technology – National Technological Institute of Mexico*, Mexico
- Jun 2025 – Nov 2025 **Graduate Certificate in Machine Learning with Python**, *CIMAT (Center for Research in Mathematics)*, Mexico
- Jan 2024 – Jun 2024 **Oracle Next Education Back-End Program**, *Alura – Oracle Next Education*, Online
- Aug 2015 – Dec 2020 **B.S. in Biochemical Engineering**, *Durango Institute of Technology – National Technological Institute of Mexico*, Mexico

Technical Skills

Programming	Python, JavaScript, Java
Systems & HPC	Linux, SLURM Job Scheduling, Distributed/HPC Computing (Discovery – NMSU, LNS-BUAP Huapáctic Supercomputer), GPU-Accelerated Computing
Frameworks & Tools	PyTorch, React, Git
Data & AI	Machine Learning, Deep Learning, Data Analysis, Data Engineering
Web Technologies	HTML, CSS, CMS Integration (WordPress)
Soft Skills	Leadership, Cross-functional Collaboration, Problem Solving, Attention to Detail
Languages	Spanish (Native), English (Full Professional Proficiency – C2)